Beyond a Single Conformer: An MD-Ensemble Tool for Dynamic 3D Descriptors of Flexible-Molecule Permeability

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1. Introduction

The landscape of modern drug discovery is rapidly expanding beyond traditional small molecules toward larger and more complex entities. Macrocycles, cyclic peptides, PROTACs, and oligonucleotides represent prominent examples of novel therapeutic modalities residing within the 'beyond rule of 5' chemical space. Efforts to modulate novel and challenging targets have substantially increased the size and complexity of drug candidates since the beginning of this century. These molecules hold the potential to target previously undruggable sites which have remained elusive to conventional small molecules. This paradigm shift is not confined to specific molecular classes, rather, it reflects a broader trend of increasing molecular size and flexibility across drug discovery. Rather than adopting a static modality, this research focuses on molecular flexibility as a shared and dynamic characteristic [1]. However, this expansion introduces formidable challenges. As molecular weight increases, both chemical space and structural complexity grow exponentially, making lead generation within this vast domain a significant hurdle. Since exploring this expanded chemical space solely through empirical experimentation is practically unfeasible due to time and cost constraints, in silico predictive tools to prescreen properties like bioavailability become indispensable [2]. Consequently, as small-molecule drug discovery ventures deeper into the complex molecules, researchers require predictive tools capable of accommodating the structural complexity, flexibility, and scale of modern therapeutic modalities.

At the core of this challenge lies the increased flexibility of these large molecules within biological environments. Compared to conventional small molecules, they undergo continuous conformational transitions in response to their surroundings. Cyclic peptides, for example, can adopt unique three-dimensional structures governed by intramolecular hydrogen-bonding networks, which effectively shield hydrophilic functional groups from the external environment. A prime example of this dynamic behavior is the 'molecular chameleon' phenomenon. Through this adaptive mechanism, macrocycles assume a more polar conformation in aqueous media but transition to a nonpolar state in low-dielectric environments. This chameleonic property underpins their capacity to simultaneously maintain high water solubility and robust membrane permeability. As a result, even when molecules exhibit identical predicted logP values, their membrane permeability coefficients can vary drastically depending on the extent of this intramolecular shielding [3]. Here, the fundamental limitation of conventional approaches become apparent. Static molecular descriptors, which are computed directly from a fixed chemical structure, condense a molecule into a single representative state. However, because QSAR frameworks rely inherently on these static properties, they frequently fail to capture the environment-dependent chameleonic behavior of macrocycles. This gap is particularly critical for macromolecules which do not exist as a single conformation but rather as a dynamic ensemble of rapidly interconverting states. In short, characterizing a flexible molecule with a static, single value fundamentally fails to capture its true physical behavior [4].

An approach to overcome this limitation is to treat molecules as dynamic ensembles. Molecular dynamics (MD) simulations generate the actual distribution of conformations that a molecule adopts in solution, enabling the calculation of 3D descriptors for each conformation – such as 3D polar surface area (3D-PSA), radius of gyration, and intramolecular hydrogen bonding. Indeed, flexibility appears to be a key determinant of membrane permeability for large macrocycles; ensembles that adopt more compact conformations with lower polar surface areas in nonpolar environments have been validated through the correlation between 3D-PSA and passive permeability. However, because predicting the 3D conformations of cyclic peptides directly from chemical structure remains a challenge, there is a clear demand for practical tools that can seamlessly generate MD-based ensembles and explore their distributions [5, 6].

In this project, we present a molecular analysis tool that addresses this demand. Our pipeline generates conformational ensembles via MD simulations using only SMILES strings as input, and extracts the distributions of 3D descriptors for interactive exploration. Through this tool, we quantitatively demonstrate that greater molecular flexibility leads to a wider variance in 3D descriptors – signifying that the representativeness of a static, single descriptor breaks down as flexibility increases. This finding directly support the necessity of ensemble-based approaches in predicting the physicochemical properties of large and flexible molecules.

2. System Overview

The developed system is an automated pipeline that streamlines the entire workflow from initial SMILES input to a comprehensive conformation report. Users are only required to provide the SMILES string of the target molecule along with specific simulation parameters – such as pH, trajectory duration, number of frames, and the desired number of clusters. Beyond this single step, all subsequent processes, including structural preparation, molecular dynamics simulation, ensemble analysis, and report generation are executed fully automatically. The key features of the system are as follows:

- **Stage-separated pipeline:** The pipeline is distinctively partitioned into three stages: structural preparation, simulation, and analysis, with each stage yielding independent intermediate outputs. Structural preparation and ensemble analysis are executed on the CPU, whereas the computationally intensive MD simulation is offloaded to the GPU. This decoupled architecture allows users to rerun only the analysis and report-generation stages without repeating the simulation, which is highly advantageous for iteratively refining reports.
- **Molecular class-branched parameterization:** Based on a rule-based classification of input molecules, peptide-like entities are routed through a protein force field pathway, while conventional small molecules are directed to a GAFF2/OpenFF pathway. This branching architecture processes diverse chemical classes within a unified workflow and is designed to accommodate future extensions into areas such as intrinsically disordered proteins (IDPs).
- **Interactive exploration report:** The system generates a self-contained HTML report for each molecule. Utilizing a single time slider, the conformation's position on the PCA/t-SNE plot, the 3D descriptor time-series data, and the 2D-projected molecular structure are synchronized to update simultaneously. By embedding only the projected coordinates into the report rather than storing heavy frame-by-frame images, the entire trajectory can be replayed within a single, lightweight file. This integration enables users to comprehensively explore the exact shape of a molecule, its location in conformational space, and its corresponding descriptor values at any given moment on a single screen.
- **Colab-based deployment for enhanced accessibility:** The AmberTools package used during the structural preparation stage requires a Linux environment and does not officially support Windows builds. To bypass this platform constraint and ensure that any researcher can utilize the system without complex environment setups, the entire pipeline is deployed as a single Google Colab notebook. By simply inputting a SMILES string and simulation parameters via a web browser, users can execute the full workflow – from preparation to report generation – leveraging the Linux-based GPU environment provided by Colab. This deployment strategy eliminates entry barriers related to local GPU availability, specific operating systems, and dependency installations, thereby significantly enhancing the system's practical usability and reproducibility.

2.1 Structure Preparation

The structural preparation stage converts a single SMILES string into a fully solvated system ready for molecular dynamics, operating entirely without GPU resources and branching its computational path based on the molecular class. To ensure reproducibility, parameters such as TIP3P water model, 12Å box buffer, and random seed are applied consistently throughout the entire process (Table 1).

Table 1. Structure Preparation Pipelines

#	Step	Small-molecule route	Peptide route	Tool
1	3D generation	ETKDGv3 multiconformer - lowest energy pick		RDKit
2	Minimization	MMFF94s geometry optimization		RDKit
3	Protonation / pH	*dimorphite-dl* at target pH,	Pdb2pqr + propka at target	dimorphite-dl/

		dominant microstate	pH	pdb2pqr
4	Partial charges	AM1-BCC (antechamber → sqm)	Library charges (ff19SB); non-standard residues flagged and logged	AmberTools
5	Parameterization	GAFF2 → mol2 + frcmod	tleap ff19SB → prmtop; explicit head-to-tail bond for macrocycles	Antechamber/ parmck2 / tleap

Input molecules are initially categorized into either conventional small molecules or peptide-like entities based on backbone amide patterns, and a review flag is recorded along with the classification rationale. In the small-molecular pathway, *dimorphite-dl* determines the dominant protonation state at the specified pH. Subsequently, RDKit ETKDG generates the initial 3D conformation, which is then minimized using the MMFF94s force field, and antechamber assigns AM1-BCC partial charges and GAFF2 atom types. In contrast, the peptide pathway utilizes the standard residue states of the ff19SB force field to build the system directly via tleap, explicitly defining head-to-tail backbone linkages for cyclic structures. For robustness, the stage incorporates automated fallback and logging. For instance, if a specific protonation state causes failure during charge calculations, the pipeline automatically retries the process once using the neutral state; meanwhile, non-standard residues (e.g., N-methylation) are detected and logged. All processing details and applied parameters are achieved in a molecule-specific manifest, enabling precise replication and stop-and-resume capability. The entire workflow is completed using only the CPU, and its outputs serve as the direct input for the subsequent GPU-accelerated simulation stage. Supplementary Table 1 shows the complete output files of this stage.

2.2 Molecular Dynamics Simulation

The prepared system is simulated in an explicit-solvent environment to generate molecule-specific conformational ensembles and frame-by-frame descriptors relevant to membrane permeability. This stage is GPU-accelerated and retains the exact charges, parameters, water model, and random seeds determined during the preceding stage to ensure reproducibility.

Following solvation (TIP3P water box and neutralizing ions), the system undergoes energy minimization, during which the pre-calculated AM1-BCC charges and parameters are strictly preserved. Equilibration is performed via NVT heating (0.1 ns) followed by NPT equilibration (0.2 ns), where position restraints on solute heavy atoms are stepwise relaxed ($10 \rightarrow 5 \rightarrow 2 \rightarrow 0$ kcal·mol^{-1}·Å^{-2}) to allow gradual structural relaxation. Subsequently, production MD is conducted at 300K and 1 bar for the trajectory duration. To optimize computational efficiency, a 4fs time step is utilized by applying hydrogen-mass repartitioning (HMR) and bond constraints, with coordinates saved every 10ps. The execution platform automatically falls back from CUDA to OpenCL, and finally to CPU, ensuring cross-environment compatibility.

The generated trajectories undergo solvent stripping and periodic boundary condition wrapping before being superimposed onto the reference structure from the preparation stage based on heavy atoms. This alignment removes global rotational and translational motions, ensuring that only internal conformational fluctuations are reflected in the subsequent analysis. For each frame, the following 3D features are computed:

- **3D-PSA:** The solvent-accessible polar surface area of N, O, and polar hydrogens, calculated using the Sharke-Rupley algorithm
- **Descriptors:** Radius of gyration (R_g), total solvent-accessible surface area (SASA), and the number of intramolecular hydrogen bonds
- **RSMD:** The root-mean-square deviation from the reference structure, used for quality control and equilibration assessment

The aligned heavy-atom coordinates are then dimensionally reduced via principal component analysis and t-SNE.

The frames are clustered via K-means to extract representative conformation for each cluster. These outputs – frame-by-frame descriptors, low-dimensional embeddings, and cluster representatives – serve as the direct input for the interactive report. Supplementary Table 2 shows the complete output files of this stage.

2.3 Report Generation

The final stage compiles the molecular MD ensemble into two distinct formats: a single feature vector per molecule and an interactive report. The ensemble, originally consisting of numerous frames, is summarized into a single row per molecule by calculating distribution statistics (mean, standard deviation, minimum, maximum and quartiles) for each descriptor, including 3D-PSA, R_g, SASA, intramolecular hydrogen bonds, and RMSD. Additionally, cluster population-weighted ensemble averages are incorporated. By merging this row with experimental labels and static 2D descriptors based on the molecular id, a consolidated table is constructed to facilitate a direct comparison between static and ensemble-based descriptors. The core analysis of this study is conducted entirely upon this table. Then, the visualization component is architected to decouple the underlying data from the viewer interface. The exporter generates a self-describing data file, which encapsulates frame-by-frame data, summary statistics, molecular topologies, pre-calculated 2D projection coordinates of the trajectory, as well as descriptor and tool versioning. A molecule-agnostic viewer renders the report for any given molecule using only this single file. Because the system stores only the projection coordinates rather than rendering heavy images, the entire trajectory can be replayed seamlessly within a single, self-contained HTML file. Supplementary Table 3 shows the complete output files of this stage.

3. Experiments

To validate what the developed system practically captures, we selected molecules spanning the entire flexibility spectrum from a Caco-2 membrane permeability dataset [7], extracted their conformational ensembles and dynamic descriptors using our pipeline, and identified the limitations of static descriptors. The following sections unfold in the order of data selection, dynamic descriptor extraction, and the analysis of static descriptor limitations.

3.1 Dataset and Molecule Selection

For the proof-of-concept endpoint, Caco-2 apparent permeability ($\log_{10}P_{app}$, cm/s) was chosen because passive permeability directly depends on a molecule's 3D shape and solvent-accessible polar surface – properties that a conformational ensemble should inherently capture more accurately than a single static structure. The initial data was sourced from the Caco-2 dataset compiled by Wang et al. (2016), which originally contained approximately 910 entries within the Therapeutic Data Commons. To curate the dataset, all molecular structures were standardized to their largest organic fragments, neutralized, and assigned canonical SMILES and InChIKeys. After removing unparsable and inorganic entries and consolidating duplicate structures by using their median label values, and 896 entries were selected. To quantitatively describe molecular flexibility, we calculated several parameters for each molecule, including the number of rotatable bonds, the Kier–Hall flexibility index (Φ), F_{sp3}, macrocyclic status, TPSA, cLogP, HBD/HBA, and the number of Rule of 5 violations. Based on the number of rotatable bonds, the molecules were categorized into flexibility tiers: rigid ($\leq$2), moderate (3–5), flexible (6–9), and very flexible (>9). The distribution across the curated 896 molecules consisted of 212 rigid, 227 moderate, 307 flexible, and 150 very flexible compounds, including 261 beyond-Ro5 (bRo5) molecules and 81 macrocycles, thereby providing a sufficiently rich dataset within the flexible and bRo5 regimes where our hypothesis is most applicable.

From this pool, a stratified and chemically diverse subset of 100 molecules was extracted for molecular dynamics simulations. Computational resources were allocated across the tiers with an intentional oversampling of the flexible tail (rigid: 15%, moderate: 25%, flexible: 30%, and very flexible: 30%), maintaining the rigid tier primarily as a negative control. Within each tier, molecules were picked using a MaxMin diversity selection based on 2048-bit Morgan fingerprints (radius 2), resulting in a subset of 15 rigid, 25 moderate, 30 flexible, and 30 very flexible molecules. To accommodate our immediate computational budget, a further subset of 40 molecules ($40 \subseteq 100$) was prioritized for MD simulation, emphasizing the flexible tail with 4 rigid, 6 moderate, 12 flexible, and 18 very flexible compounds. This practical subset spans a rotatable bond count of 2–18 and a Kier Φ index of 2.3–20.9, while encompassing 3 macrocycles and 18 bRo5 compounds. Ultimately, 37 of these molecules successfully completed the production MD runs to form the final analysis dataset.

3.2 Extracting Conformational Ensembles and Dynamic Descriptors

The automated pipeline was executed for each of the 37 curated molecules (yielding a 10 ns trajectory per molecule) to obtain explicit-solvent conformational ensembles and frame-by-frame 3D descriptors. Given our core hypothesis that more flexible molecules are inherently more challenging to represent using a single static value, we quantified this phenomenon by using the variance of each descriptor across individual frames as a metric for flexibility. Specifically, the standard deviation of descriptor values across all production-phase trajectory frames reflects the intrinsic uncertainty associated with representing that particular molecule by a single static value. To ensure rigorous quantification, several methodological choices were implemented. First, the non-equilibrium initial segment of each trajectory was identified through temperature, density, and energy stabilization, and subsequently excluded from the variance calculations. This step ensured that our metrics exclusively captured the true conformational flexibility rather than artifacts from the equilibration process. Second, the variance was computed directly from the entire frame distribution rather than from fixed clusters. Approaches that enforce a fixed number of clusters can artificially segment even rigid molecules, thereby distorting the signal corresponding to the actual number of conformational states and rendering them unsuitable for quantifying flexibility. Third, because the root-mean-square deviation (RMSD) is a relative value dependent on an arbitrary reference structure (e.g., at 0 ns) and thus confounds direct inter-molecular comparisons, we relied on reference-independent individual frame variances as our robust metrics for flexibility. As a result, we observed a clear trend where increasing molecular flexibility corresponds to a larger descriptor variance. In Figure 1, strong positive correlations emerged between the molecular flexibility indicators—namely, the number of rotatable bonds and the Kier–Hall flexibility index (Φ)—and the individual frame variances of the 3D descriptors. The variance of 3D-PSA exhibited a Spearman correlation of $\rho = 0.51$ ($p = 0.0013$) with the number of rotatable bonds, and $\rho = 0.55$ ($p = 0.0004$) with the Kier Φ index. Similarly, the variance of the radius of gyration (R_g) demonstrated a consistent trend, yielding correlations of $\rho = 0.55$ ($p = 0.0004$) and $\rho = 0.50$ ($p = 0.0015$), respectively ($n = 37$ for all cases). Visually, rigid molecules (colored blue) cluster tightly within regions of low variance, whereas highly flexible molecules (colored red) scatter broadly toward significantly wider variance distributions.

3.3 Where Static Descriptors Fail

While Section 3.2 summarized descriptor variance into a single quantitative metric, we here examine the actual shapes of these variances and identify precisely when static descriptors fail by analyzing the full distributions. Specifically, the 37 completed molecules were ordered by their Kier Φ index—progressing from rigid at the bottom to flexible at the top—to concurrently map each molecule's molecular dynamics (MD) ensemble 3D-PSA distribution via violin plots against its single static structure-derived 3D-PSA value represented by diamonds (Figure 2). As expected, the shape of the distributions visibly broadens in tandem with increasing molecular flexibility. The violin plots for rigid molecules (bottom, shaded blue) remain highly narrow, indicating that their 3D-PSA values undergo negligible fluctuations throughout the simulation. Moving upward toward the flexible regimes (top, shaded red), the violin plots expand significantly, revealing that a single molecule dynamically transitions among various conformations with vastly differing polar surface areas. This morphological expansion visually reaffirms the variance-flexibility correlation established in Section 3.2. Furthermore, superimposing the static 3D-PSA value of each molecule onto its corresponding ensemble distribution demonstrates a frequent divergence, where static values falling outside the 95% confidence interval of the ensemble are highlighted with red borders. For rigid molecules, the static values generally reside within or at the immediate periphery of the narrow distributions, meaning a single static point serves as a reasonably fair surrogate for the ensemble. In stark contrast, the static values for highly flexible molecules frequently deviate far from the distributional center. For instance, the ensemble of Ac-Trp-Ala-Gly-Gly-Lys peptide (Φ = 14.5) exhibits a wide 3D-PSA distribution that significantly exceeds 300 $Å^2$ whereas its corresponding static structure provides only a single point that cannot capture the full range of the ensemble. These observations underscores a fundamental, first-principles limitation: static descriptors are mathematically and physically incapable of capturing the extensive range of structural states that a flexible molecule actually samples.

4. Discussion

In this study, we established an automated pipeline that seamlessly generates explicit-solvent molecular dynamics (MD) ensembles directly from SMILES inputs, extracts the full distributions of 3D descriptors, and enables interactive exploration of the resulting data. Applying this framework to 37 benchmark molecules spanning the entire flexibility spectrum revealed that increasing molecular flexibility strongly correlates with a larger individual frame variance in 3D descriptors, particularly 3D-PSA (Spearman $\rho \approx 0.5$). Furthermore, we demonstrated that for highly flexible molecules, descriptor values derived from a single static structure frequently fail to represent the broader ensemble distribution. These empirical findings substantiate a fundamental, first-principles limitation of static single-conformation descriptors in predicting the physicochemical properties of large, flexible molecules, underscores the absolute necessity of ensemble-based computational approaches.

Crucially, the fact that the correlation between molecular flexibility and descriptor variance is moderate ($\rho \approx 0.5$) rather than perfect, which might superficially seem like a limitation, actually serves as a compelling justification for the necessity of our pipeline. If formal flexibility indices could perfectly predict descriptor variance ($\rho \approx 1.0$), performing computationally expensive MD simulations would be redundant, as simply counting the number of rotatable bonds would suffice. This moderate correlation implies that while macro-level flexibility dictates the rough boundaries or overall propensity, the actual realized variance is highly molecule-specific and can only be uncovered through direct simulation. For instance, molecules sharing the exact same number of rotatable bonds (e.g., $N_{rot} = 8$) exhibited 3D-PSA variances that diverged drastically, ranging from 3 to 20Å^2. The number of rotatable bonds represents merely a potential or latent flexibility; the actual conformational diversity manifested under biological or explicit-solvent environments heavily hinges upon the unique structural context of the molecule. It is precisely this emergent, non-trivial structural information that our ensemble simulation captures.

In the same vein, this trend operates as a general propensity rather than an absolute mathematical law. The relationship between flexibility and descriptor variance is inherently non-monotonic: certain highly flexible molecules exhibit relatively narrow distributions—often owing to strong, stable intramolecular interactions that lock the conformation—whereas some moderately flexible molecules display remarkably wide distributions. This non-monotonicity is not a flaw in our model, but rather a core physical reality. If flexibility deterministically governed variance by a strict law, simulation would merely be an unnecessary duplication of static topology rules. The predictability of the macro-trend, coupled with the unpredictability of individual molecular behavior, is precisely what underscores and justifies the utility of explicit ensemble calculations.

Finally, we emphasize that this ensemble-based framework is not intended to dogmatically replace static descriptors, but rather to delineate exactly when static metrics remain reliable. For rigid molecules, static values serve as highly accurate surrogates since their distributions are tightly bounded, rendering explicit simulations unnecessary. The true value of ensemble calculations is concentrated within the highly flexible regimes—the precise domain where static points fail. Consequently, rather than being applied indiscriminately to all chemical structures, our tool functions as a critical diagnostic gatekeeper: it determines whether a molecule can be safely simplified into a single static value or if its full distribution must be mapped, subsequently delivering that exact distribution when necessary.

The practical utility of the developed pipeline is fully realized through its deployment strategy. The entire workflow—encompassing SMILES and simulation condition inputs, structure preparation, MD simulations, ensemble analyses, and interactive report generation—has been packaged and made publicly available as a single Google Colab notebook [8] and Github repository (https://github.com/errory230/CV_LifeSciences). Without the need for local GPUs, specific operating systems, or complex dependency installations, users can execute the entire process within a browser by simply opening the notebook and inputting a molecule's SMILES string, leveraging the Linux-based GPU environment provided by Colab. Consequently, researchers lacking specialized computational chemistry infrastructure can instantly retrieve conformational ensembles and 3D descriptor distributions for any given molecule. Because the generated reports operate independently as self-contained files, they facilitate seamless sharing and robust reproducibility. This high level of accessibility elevates the pipeline beyond a mere methodological demonstration into a highly practical, ready-to-use research tool.

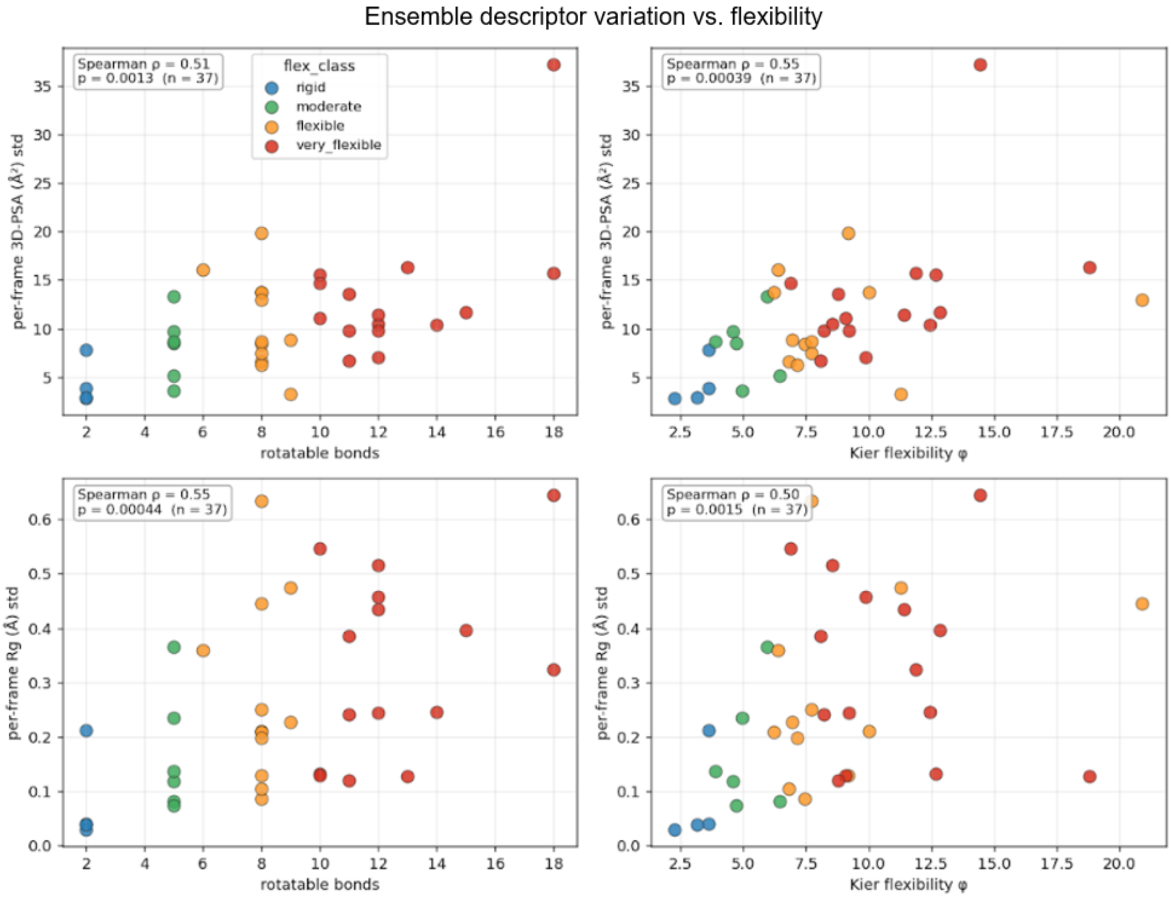

Figure 1. Ensemble descriptor variation vs. molecular flexibility (n=37). Per-frame standard deviation of 3D-PSA (top) and radius of gyration (bottom) plotted against rotatable-bond count (left) and Kier-Hall flexibility (right). All four relationships show significant positive correlation (Spearman ρ = 0.50-55, $p < 0.002$). Colors denote flexibility classes; rigid molecules serve as a negative control.

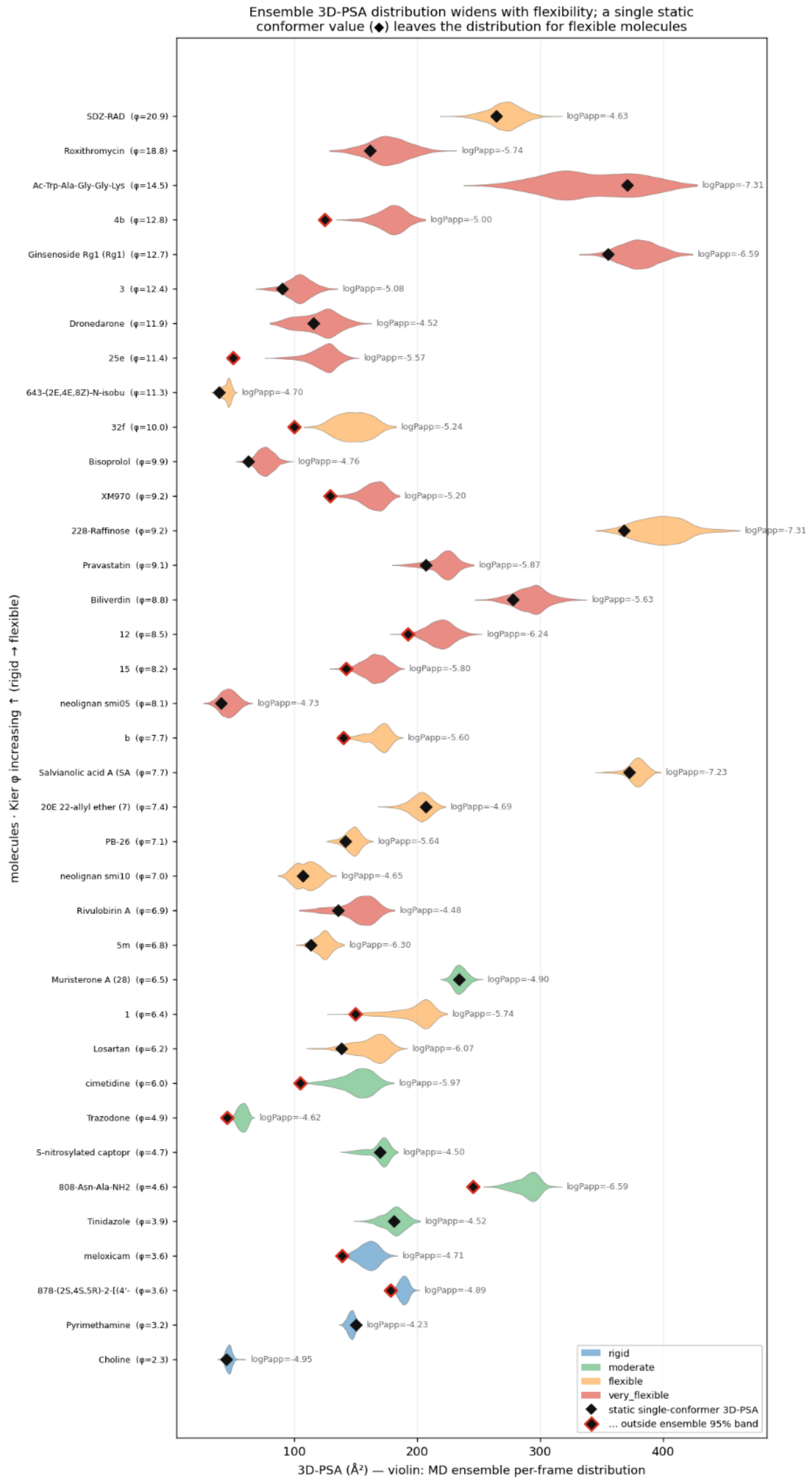

Figure 2. Ensemble 3D-PSA distributions for all 37 molecules, ordered by Kier-Hall flexibility. Violin shows the MD ensemble per-frame 3D-PSA distribution. Diamonds mark the 3D-PSA of the single static conformer. Red-outlined diamonds fall outside the ensemble's 95% band.

Referecne

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Supplementary Table 1. Output files – Structure Preparation Step

File	Meaning	Format
ligand.mol2	AM1-BCC-charged, GAFF2-typed structure	MOL2
ligand.frcmod	Supplementary GAFF2 parameters	frcmod
ligand_gas.prmtop / .rst7	Gas-phase topology + coordinates (validation)	AMBER
ligand_ref.pdb	Reference single conformer (the "static" structure)	PDB
ligand.sdf	Minimized conformer	SDF
leap_solvate.in	tleap recipe → TIP3P box, neutralizing ions, 12 Å padding	tleap input
manifest.json	Full provenance: class, protonation state, net charge, force field, tool versions, per-step status	JSON

Supplementary Table 2. Output files – MDS step

File	Meaning	Format	Use
solvation/ligand_solv.prmtop	Solvated topology (atoms, bonds, charges)	AMBER	Trajectory parsing
trajectory.dcd	All production frames (solute + water)	DCD	3D-descriptor recomputation source
analysis/metrics.csv	Per-frame RMSD, 3D-PSA, PC1/2, tSNE1/2, cluster	CSV	Feature-vector aggregation
analysis/ensemble_summary.json	Descriptor mean/std/min/max, cluster populations, PCA variance	JSON	Feature
analysis/clusterNN.pdb	Representative structure per cluster (solute only)	PDB	Weighted-ensemble representative
statedata.csv	Temperature / density /	CSV	QC (equilibration check)

	energy time series		
run_manifest.json	prep + MD parameters, platform, frame count	JSON	Reproducibility / provenance

Supplementary Table 3. Output files – Ensemble Analysis Step

Artifact	Meaning
features.csv	one QSAR feature row per molecule (ensemble distribution stats + weighted means)
feature_detail.json	per-molecule summary + per-cluster means + descriptor definitions
explore_<mol_id>.html	self-contained explorer (inline CSS/JS, base64 — no external files/CDN)
index.html	hub catalog across all molecules